TP04: Containerization with Docker

Big Data Analytics | Master 2 IA & Data Science | Docker Hub image: alaaguedda/tp04-higgs:1.0

Task 4: Docker Compose

What problem does Docker Compose solve?

1. It describes the whole application (build, volumes, network) in **one compose.yaml** file.
2. One command starts it the same way on any machine, instead of many long docker run commands.

docker run vs docker compose up

3. **docker run** starts one container and every option is typed by hand each time.
4. **docker compose up** reads compose.yaml, builds, creates volumes and starts all services.

How to add a database service?

5. Add a second service (**image: postgres:16**) with a named volume for its data.
6. The app reaches it by the service name (host db); depends_on starts the database first.

Task 5: Report

What is Docker and why is it useful in data science?

7. Docker packages an application with its libraries into a container that **runs the same everywhere** .
8. It gives reproducible results, no version conflicts and easy sharing through Docker Hub.

Image vs container

9. An **image** is a read-only template (the recipe); a **container** is a running instance of it.
10. One image can create many containers, like one class creates many objects (we ran three).

Dockerfile instructions

11. **FROM** base image, **WORKDIR** working folder, **COPY** copies files from the computer into the image.
12. **RUN** executes a command at build time (pip install); **CMD** is the command run when the container starts.